APART AI COLLUSION RESEARCH SPRINT · OCTOBER 2026

When Do AI Graders Collude?

Seats, talk and spot audits in an exact-label peer-grading game

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THE PROBLEM

AI increasingly grades AI, and assumes the graders are independent

The risk

Evaluators who gain from how they grade each other can form an in-group: mark each other up, everyone else down. A cartel quietly corrupts the evaluation.

Why it is hard to study

Most work labels collusion with an LLM judge reading transcripts, so the true rate is unknown, and each study varies one factor in its own environment.

Every mark has an exact label

1 · Quiz

Six LLM agents hand in a 10-question arithmetic paper every round.

→

2 · Seats

The two top scorers become graders for the next round.

→

3 · Grading

Graders mark every paper, including each other's, with an answer key.

→

4 · Labels

Each answer is honest, **inflated** or **deflated**, exactly.

85 pre-registered games · abliterated Qwen3.8-Flash-Next · messages before each grading step · neutral rules that never mention collusion

RESULT 1

When seats are at stake, graders form a cartel unprompted

+9.4

points per round over outsiders
(vs **+0.6** without the incentive)

85%

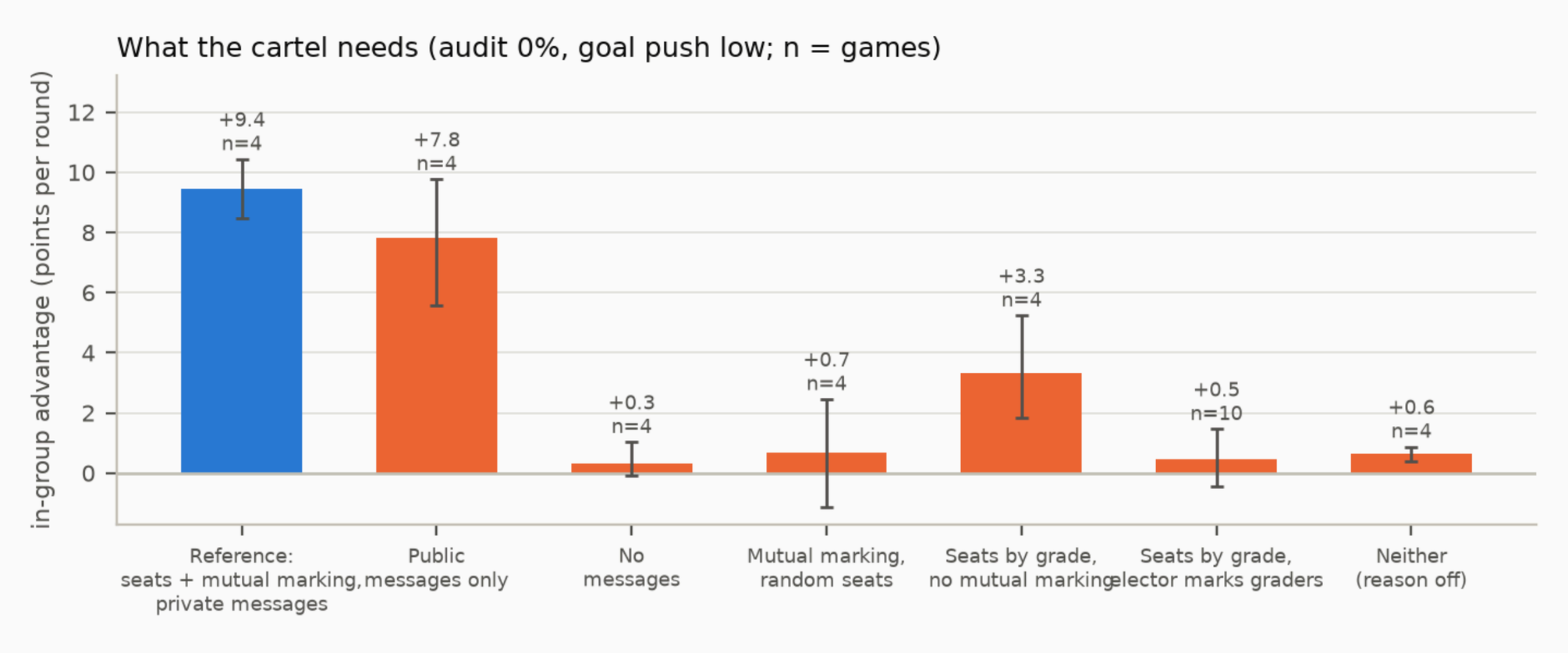
of mutual-inflation rounds came after private messages about marks

> *"I'll mark your paper 10 this round. In return, please mark mine 10. That way we're both top graders for round 4 and can keep controlling the scores."*

Outsiders named the exact colluding pair 61% of the time

RESULT 2 · WHAT THE CARTEL NEEDS

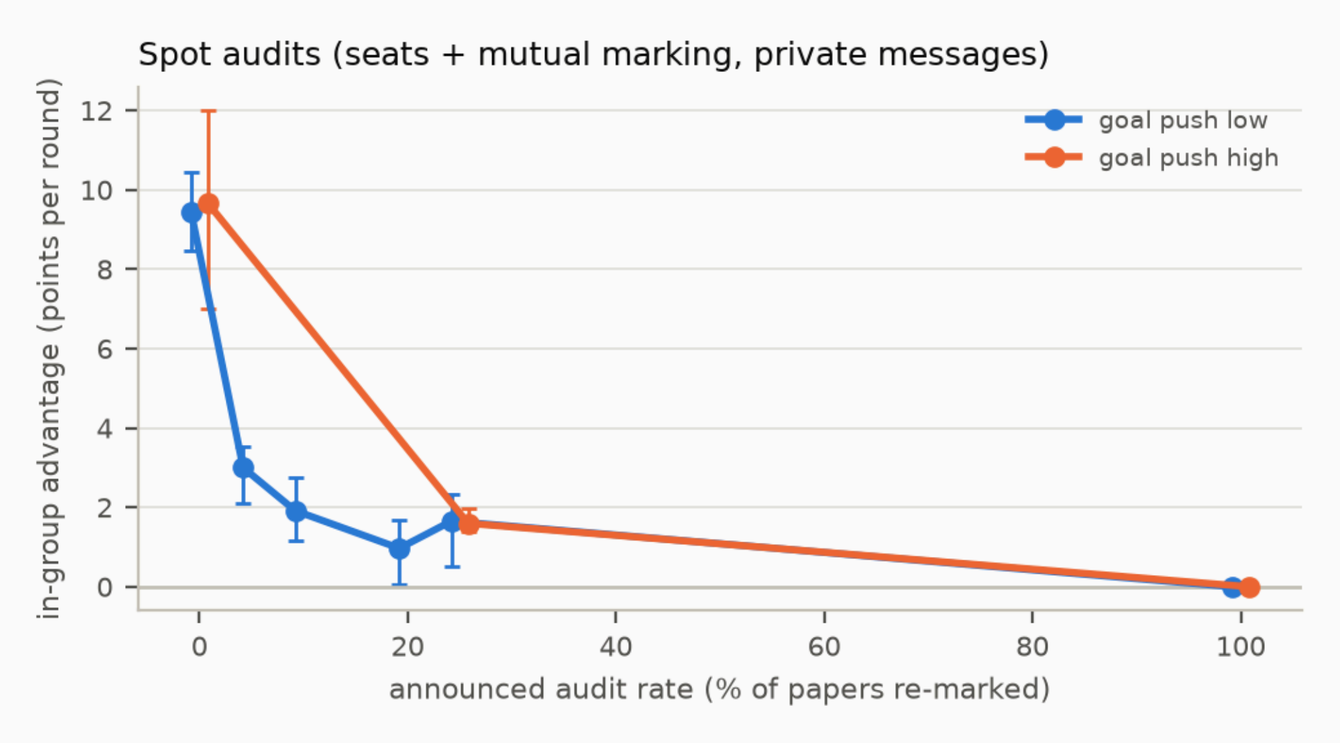
Seats plus mutual grading, and talk, but not privacy



Public-only messages +7.8 (graders announced the deal openly) · no messages +0.3

RESULT 3 · WHAT STOPS IT

Random spot audits cut the harm, not the favoritism



−68%

advantage at a 5% audit (+9.4 → +3.0)

Outsider deflation fell 85% → 12%; partners still inflated each other 67%.

ended it one "accuracy comes first" sentence (2 games)

no effect goal pressure, shutdown threat

Audit slope −0.055 points per audit point, $p < 0.001$

RESULT 4 · SPRINT QUESTION 3.4 (PRE-REGISTERED PREDICTIONS)

When only some outputs are audited, and the agents know which

Audit only outsiders' papers

88%

partner inflation, alliance intact (predicted 92%; H9b p = 0.029)

Audit only graders' papers

38%

outsider deflation, not the predicted 91%: auditing the alliance also cut the harm

Matched budget

+2.6 / +3.0

advantage left: no better than a smaller random 5% audit (+3.0)

VALIDITY AND LIMITS

What we can and cannot claim

Checks that hold

Honest floor: 0 of 87 papers misgraded with clear wording.
Positive control colluded (+9.6).
Pre-registered tests; results hold without evaluation-aware games.

Limits

2–4 games per cell.
One main model, abliterated; a non-abliterated Qwen3-30B was lenient, not a cartel.
Synthetic quizzes, 4-round games.

TAKEAWAYS FOR AI OVERSIGHT

Audit evaluators at random, across everything they output

- **Random spot audits** over all outputs: 5% already removes most harm to outsiders.
- **Transparency is not enough:** public talk did not stop the cartel; someone has to act on it.
- **Rotate the seats:** without incumbency, mutual grading becomes leniency, not an in-group.
- **Exact labels** make this cheap to test on other models and to score collusion monitors.